

The Semantics of Negation

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Abstract

We classify negative elements in natural language into five broad categories: standard negators, negative correlative conjunctions, negative indefinites, other types of negative elements, and squattive negators. We discuss a number of criteria by which to diagnose negation, such as the universally negative reading of *or* in the presence of negation (based on De Morgan's Law), the licensing of Negative Polarity Items (NPIs), the interaction with modality, and the syntactic effects of negation. We conclude that the diagnostics point to the conclusion that some negative morphemes are only semantically negative, while others are negative both syntactically and semantically.

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1 Introduction

Classical logic has only one way of negating a proposition, by means of the negative operator, which has one fixed meaning, i.e. reversing the truth value of the proposition it combines with.¹

- (1) p
 $\neg p$

However, morphologically speaking, negation takes many different forms and gets packaged in intricate ways with other operators and/or other functional categories. Moreover, not all of these seemingly negative forms semantically correspond to simple truth value reversers. In this chapter, we shall therefore start out in (Sect. 2) by examining a plethora of forms that negation can take in natural language: standard negators (Sect. 2.1), negative indefinites (including negative quantifiers and negative concord items; Sect. 2.2), negative correlative conjunctions (Sect. 2.3), squative negation (Sect. 2.4), and other negative words and expressions (Sect. 2.5). Next we will test these different forms against a set of diagnostic tests, which on the one hand seem suitable for diagnosing the presence of a semantic/conceptual or logical negation, and which on the other hand diagnose something which seems to be more syntactic in nature, and that we will refer to by the term ‘negative scope’ (Sect. 3). We will conclude that while some negative morphemes are only semantically negative, others are negative both syntactically and semantically (see also Zeijlstra, 2004b, 2008 for a similar idea). Moreover, if a morpheme is syntactically negative, the level of embedding of the negative feature and the internal morphosyntactic make-up of the negative marker are relevant to its scope in the sentence.

2 Negative elements in Natural Language

2.1 Standard negators

The most common negator across the languages of the world is the standard negator. With the term ‘standard negator’ Miestamo (2005) refers to the negator that is used to produce the simple negation of a declarative sentence, as in the English example (2b) below.

- (2) a. Alissa went to school today.
 b. Alissa did **not** go to school today.

In English, the standard negator is the negative adverb *not*. All languages have a morpheme or word that can make a sentence negative in this way (Dryer, 2013; Horn, 2001b), and we will look at the morphology of standard negators in Sect. 2.1.2. Before we do that, we want to briefly discuss the core meaning of standard negators, in particular, the distinction between internal and external negation.

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2.1.1 External vs. internal negation

Based on the crucial role of a standard negator in the languages of the world one would assume that it is the obvious counterpart to the negative operator of classical logic. However, this is not straightforwardly true. Looking at (2b), at the surface level it is not immediately clear that the negation has the entire proposition in its scope, as is the case for the negative operator from classical logic (see (1) above). More concretely, in (2b) the subject and the tensed auxiliary do not seem to be included in the surface scope of the negator *not*. This points us to an important difference between the traditional operator from logic and the standard negator across the languages of the world: while the first one is an external negator, representing the meaning ‘It is not the case that p ’, the regular standard negator is an internal negator, scoping over the tensed or untensed predicate. More importantly, it has been claimed that there are no languages in the world that have a dedicated morpheme for external negation (Horn, 2001b). That does not mean of course that there are no means to do external negation: negative clefts or negative preposed DPs, for instance, can give rise to external negation. In fact, it has been proposed that Jewish Babylonean Aramaic is a language with a dedicated grammaticalised external negator (Bar-Asher Siegal, 2015; Bar-Asher Siegal & De Clercq, 2019, see also the discussion of Boumaa Fijian in Sect. 2.1.2 below). Moreover, while the standard negator is clearly not a straightforward counterpart of the logical negator, it very often also gives rise to external negation. The meaning of (2b) can, in fact, be paraphrased as in (3).

- (3) It is not the case that Alissa went to school today.

However, when a scope bearing element like the quantifier *many* occupies the subject position (or is contained in the subject), as in (4a), the meaning of the sentence with the regular standard negator (4a) does not correspond to the meaning of the sentence with the external negator (or negative cleft in this case), illustrated in (4b) (Jackendoff, 1972, p. 326).²

- | | | |
|-----|---|---------------|
| (4) | a. Many of the arrows didn’t hit the target | MANY > $\neg$ |
| | b. It is not the case that many of the arrows hit the target. | $\neg$ > MANY |

The reading of the external negation, i.e. the one in (4b), can also be achieved by adding the negator to the subject, as in (5) (Jackendoff, 1972, p. 326):

- (5) Not many of the arrows hit the target.

The difference seems first and foremost to be one of scope, as is indicated in (4). It may be brought out by the continuation ‘but many of them did hit the target’, as shown in (6).³

²Not all quantifiers behave the same in this respect: universal quantifiers in subject position are scope-ambiguous with internal negation (e.g. *All that glitters is not gold*; Jackendoff, 1972, 352ff; Horn, 2001b, 226ff). However, here too there is a contrast with external negation, which causes the ambiguity to disappear: *It is not the case that all that glitters is gold*.

³One could view external and internal negation as a consequence of scopal differences for one and the same $\neg$, but one could also treat them as instances of two different (semantic) types of negation, with a different lexical semantics. Within systems that adopt a different semantics for internal ($\neg p$) and external ($\neg p$) negators, the main difference is that the external negator is presupposition cancelling, while the internal negator is presupposition preserving (see for instance Bar-Asher Siegal, 2015, forthcoming; Bar-Asher Siegal & De Clercq, 2019; Horn, 2001b; Ladusaw, 1980; Schwarz & Bhatt, 2006). Horn (2001b) refers to those who recognize a different external and internal negator as ambiguists, and to those who adopt only one negative operator with one meaning and one scope as monoguists (see among many others Gazdar, 1979). We refer the reader to the literature referred to for more discussion of the

While (4a) could be continued as in (6), (4b) and (5) could not, as illustrated in (7a) and (7b), respectively.

- (6) Many of the arrows didn't hit the target, but many did.
- (7) a. #Not many of the arrows hit the target, but many did.
b. #It is not the case that many of the arrows hit the target, but many did.

This discussion shows that while standard negators very often indeed correspond to the external negator from classical logic, they cannot really be equated with it. Standard negators are always internal negators with the capacity to extend their scope to that of an external negator, when there are no intervening operators. This is an important point to keep in mind while we discuss the morphology of standard negators in the next section.

2.1.2 A morphological classification

Dryer (2013) gives a typological overview of the different ways in which standard negators can be realised.⁴ He discusses five different types of standard negators. The first type is affixal negation, where the standard negator is an affix on the verb, as in Kolyma Yukaghir, spoken in Siberia. An example is given in (8):

- (8) *Kolyma Yukaghir* (Siberia)
met numö-ge el-jaqa-te-je.
1SG house-LOC NEG-achieve-FUT-INTR.1SG
'I will not reach the house.'

Standard negators of the second type have what Dryer calls a 'negative particle', i.e. a separate word that makes the sentence negative. Musgu is such a language:

- (9) *Musgu* (Chadic, Afro-Asiatic; Cameroon)
à sǝǝ cǝcǝbǝ pǝy.
3SG.M know jackal NEG
'He didn't see the jackal.'

The third type uses a negative auxiliary: this is a negative word that shows properties of a verb, such as showing agreement with the subject. Finnish has such a special negative auxiliary:

- (10) *Finnish* (Finnic; Uralic)
e-n syö-nyt omena-a.
NEG-1SG eat-PTCP apple-PART
'I didn't eat an apple.'

In Grebo we see something similar to Finnish. In the affirmative sentence (11a) the verb *du* 'to stamp' gets a special tense suffix *-da*², which refers to a 'past before yesterday'. In the negative sentence the same tense suffix appears on the negative word *yi*²¹.

- (11) *Grebo* (Kru, Niger-Congo; Liberia)

distinction between external and internal negation.

⁴All the examples in this section are taken from Dryer (2013), except where otherwise indicated. Dryer's overview does not include allomorphs of the standard negator in non-declarative clauses (see De Clercq 2020, 2025).

- a. ne¹ du¹-da² bla⁴.
1SG.SUBJ pound-PST rice
'I pounded rice before yesterday.'
- b. ne¹ yi²¹-da² bla⁴ du¹ mɔle.
1SG.SUBJ NEG-PST rice pound Monday
'I did not pound rice on Monday.'

There also exist standard negators for which it is not clear *a priori* whether they are a word (and thus belong to the second type), or a verb (which would make them cases of the third type). The reason for this is that there is no agreement morphology in the language in question.

- (12) *Maori* (Tahitic; Austronesian; New Zealand)
kaahore taatou e haere ana aapoopoo.
NEG 1PL.INCL T/A move T/A tomorrow
'We are not going tomorrow.'

In Boumaa Fijian, although there is no agreement morphology, the standard negator does appear to be more of a verb, with the sentence to be negated a subordinate clause that functions as the subject of the negative verb:

- (13) *Boumaa Fijian* (Oceanic)
e sega [ni la'o o Jone].
3SG NEG COMP go ART John
'John is not going.' (literally: 'It is not the case that John is going.')

Dryer calls the last type in his classification 'double negation', to describe a situation where two discontinuous morphemes correspond with one logical negation. In Izi these morphemes are a prefix and a suffix on the verb.

- (14) *Izi* (Igboid, Niger-Congo; Nigeria)
ó tó-òmé-ǎú ré.
3SG NEG-do-NEG well
'He does not do well.'

In Ma one of the negative morphemes is an affix on the verb (*tá*), and the other a free morpheme (*nyɔ̌*).

- (15) *Ma* (Adamawa-Ubangi, Niger-Congo; Democratic Republic of Congo)
tá-mù-sùbù-li nóṅgbó nyɔ̌.
NEG-1SG-eat-PST meat NEG.1SG
'I did not eat meat.'

However, we will use the term 'bipartite negation' to describe this type, because two morphemes are needed to make the sentence negative, while reserving the term 'double negation' for cases where there are actually two semantic negations.

Double negation arises in an English sentence like the following, with two occurrences of the standard negator:

- (16) We did **not not** talk to them.

Since each of the two negators contributes a logical negation, the combination of them yields a double negation, with the two negations cancelling each other out and yielding an affirmation, according to the law of double negation (*‘duplex negatio affirmat’*).

$$(17) \quad \neg\neg p \leftrightarrow p$$

In many languages with bipartite negation, one of the morphemes is optional, and when there is only one negative morpheme, it is usually the free morpheme. This is the case in West Flemish for instance (18) (Haegeman, 1995, p. 120).

- (18) da Valère woarschijnlijk **nie** styf drunke (**en**) was
 that Valère probably not very drunk NEG=was
 ‘that Valère probably wasn’t very drunk.’

Observe that although we gloss *en* as NEG, *en* by itself is not sufficient to make a sentence negative in current varieties of (West) Flemish: it requires the simultaneous presence of *nie*. *En* on the other hand can be easily dropped and has been argued to have become a discourse marker rather than a polarity marker (Breitbarth & Haegeman, 2014). This optionality in (18) illustrates one step in a diachronic process for negation that is well-known as Jespersen Cycle; see Jespersen (1924, pp. 335–6), Dahl (1979, p. 88) for coinage of the term, and Willis et al. (2013) for an overview. This cycle involves an originally negative element being strengthened by a second element, which at first is optional, but becomes compulsory over time, resulting in a bipartite negation. The second element then becomes the bearer of the negative meaning. This weakens the original first element and makes it disappear over time.⁵ This pattern of diachronic development is seen in many different languages. A schematic representation of it is given in Figure 1 (from Hoeksema, 2009).⁶

⁵There are different proposals in the literature concerning the real trigger for the change. While Jespersen (1917) argued that it is due to the weakening of the original negator that a new element enters the scene, others have argued that it is the introduction of a new emphatic element that weakens the original negator (Kiparsky & Condoravdi, 2006; Meillet, 1921; Mosegaard Hansen, 2013, pp. 51–53).

⁶Note that Hoeksema’s diagram uses the term ‘double negation’, which we refer to as ‘bipartite negation.’

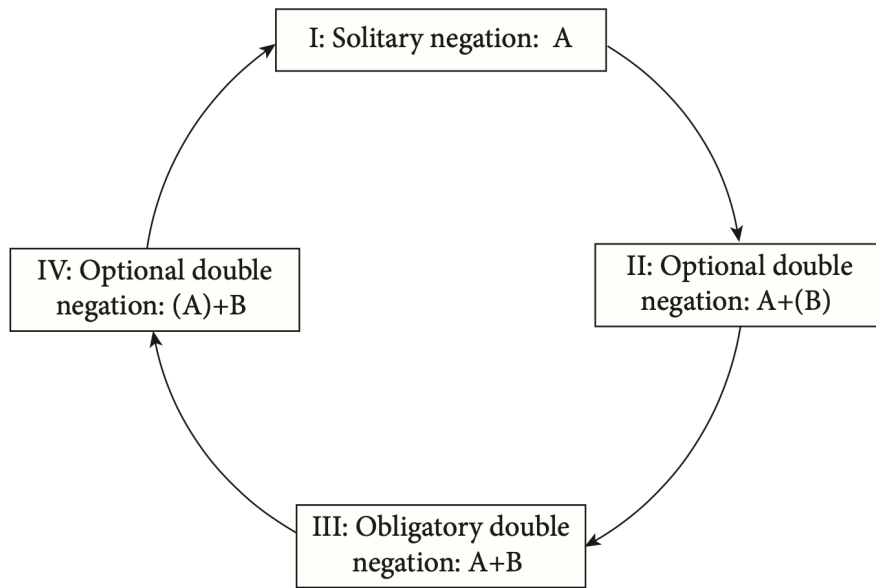


Figure 1: The Jespersen Cycle

We see the same phenomenon in written French, which reflects an older stage of the language: the proclitic element *ne* on the verb combines with a second negative element *pas* (19). Just like in West Flemish the *ne* can be dropped and is actually mostly dropped in colloquial French.

- (19) Je **ne** bois **pas** de bière.
 I NEG=drink not of beer
 'I don't drink beer.'

The crucial point for this chapter is that in the stage when the two elements need to co-occur, there are in essence two different morphemes responsible for the lexicalisation of the single logical negation $\neg$. This could suggest that $\neg$ needs to be decomposed, or that $\neg$ is underlyingly always associated with another element. As for the first possibility, to our knowledge no one has tried to decompose the operator $\neg$ in semantic subcomponents that together give rise to the meaning associated with $\neg$. Most proposals in the literature take a version of the second approach, which deals with bipartite negation as a temporary morphosyntactic strengthening of the negator. Just like Kiparsky and Condoravdi's (2006) semantic/pragmatic account, most morphosyntactic proposals are in essence asymmetric, putting the semantic negativity in one of the two elements, with the other element dependent on it or reinforcing the negator; see amongst others Zeijlstra (2008); Rizzi and Roberts (1996, p. 76); Rowlett (1998, p. 28); Roberts and Roussou (2003, pp. 154–155); Roberts (2007, pp. 64–81). The problem with all of these accounts is that the co-occurrence of the two morphemes in stage III of the cycle goes without explanation, i.e. while most of these proposals explain how the non-negative element can be licensed or accounted for, they do not explain why the inherently negative element cannot occur on its own. While descriptively adequate, these accounts give no deeper insight into the reasons for the presence of the two morphemes in stage III of the cycle. The syntactic accounts of Poletto (2008) and De Clercq (2013, 2020) try to capture this obligatory co-occurrence of French *ne* and *pas* by proposing that the marker responsible for

sentence negation or wide scope negation not only needs to lexicalise NEG, but also a fixed set of other tense or scope related features. We refer the reader to the literature mentioned for more details on bipartite negation.

2.2 Negative indefinites

The previous section has discussed standard negators. In this section, we turn to a second type of negator, the negative indefinites. They derive their name from the fact that, semantically, they contain a negation $\neg$ and an existential quantifier $\exists$, with the negation taking scope over the quantifier and giving rise to a universally negative meaning ($\neg\exists$) (Haspelmath et al., 2005; Penka, 2011). We will look at two types of negative indefinites, those that always give rise to negation on their own, the so-called negative quantifiers (Zeijlstra, 2020; Sect. 2.2.1), and those that do not always or hardly ever give rise to negation on their own, the so-called negative concord items (Sect. 2.2.2).

2.2.1 Negative quantifiers

Negative quantifiers are a subtype of negative indefinites, characterised by the fact that they have semantically negative force on their own. Examples are given in (20).

- (20)
- a. **No-one** knew the answer.
 - b. There was **nothing** left of the delicious dessert.
 - c. He **never** watches football.
 - d. They are welcome **nowhere**.
 - e. **No** student knew the answer.

In English, they formally consist of *n-* and a quantificational pronoun, adverb, or *wh-word* like *one*, *ever*, or *where*.⁷

In addition to $\neg$, these negative indefinites also lexicalise an existential quantifier $\exists$. This $\exists$ corresponds to the quantificational pronoun (e.g. *someone*) or adverb (*ever*, *somewhere*) contained in these negative indefinites. In other words, the meaning of these words can be understood as a combination of the logical symbols $\neg$ and $\exists$. This decomposition is shown schematically in (21) (see Collins & Postal, 2014, for an argument that this decomposition is also syntactic).

(21)

	$\neg$	$\exists$
no-one	= not	someone
no-thing	= not	something
n-ever	= not	ever
no-where	= not	somewhere
no N	= not	a N

Somewhat informally, then, and considering what we said about the scope of $\neg$ in classical logic, i.e. it being an external negator, we could represent the decomposition of the negative words in the examples of (20) as in (22).

⁷Note that the meaning of a negative quantifier can also be expressed by means of an standard negator and an indefinite (compare *There is nobody there* with *There isn't anybody there*). This is the case for instance in Mandarin Chinese; see Paul (2021) for further discussion.

- (22) a. $\neg$ (Someone knew the answer)
 b. $\neg$ (There was **something** left of the delicious dessert)
 c. $\neg$ (He **sometimes** watches football)
 d. $\neg$ (They are **welcome** somewhere)
 e. $\neg$ (A student knew the answer)

These representations are informal not only in their notation, but also in the fact that they show all these negations as having an external negation, i.e. with the negation having the entire proposition in its scope. As we shall show when discussing the syntactic effects of negation (Sect. 3.4 below), negative indefinites do not always take this wide scope.

Since negative quantifiers contribute a logical negation by themselves, the combination of a negative quantifier with a standard negator will yield a double negation, as in the English example in (23), with the standard negator and a negative quantifier in object position:

- (23) We did **not** invite **no-one**.

The resulting reading is an affirmative one, i.e. *not no-one* ($\neg\neg\exists$) means *someone* ($\exists$).

2.2.2 Negative concord items

A second type of negative indefinites looks like the negative quantifiers, but differs from them in that they do not always give rise to negation on their own, but require the additional presence of a negative morpheme, which can be the standard negator, or another negative indefinite. An example from Italian is given in (24).

- (24) **Non** ha visto **niente**.
 not have.3SG seen nothing
 ‘He has seen nothing.’

The negative indefinite *niente* ‘nothing’ requires the presence of the standard negator *non*. In the case of negative quantifiers discussed in the previous section, this combination resulted in a double negation, i.e. an affirmative. In contrast, in the case of (24) there is only one semantic negation. This phenomenon is referred to as negative concord, i.e. two (or more) negative morphemes agree with each other to only realise one semantic negation (see Giannakidou, 2000; Haegeman, 1995; Jespersen, 1922; Van der Auwera & Gianollo, 2025; Zeijlstra, 2004a, and many others). Correspondingly, negative indefinites that display concord properties are referred to as ‘negative concord items’. An alternative term used in the literature is that of *n-word* (Giannakidou, 2006; Laka, 1990). Negative concord resembles bipartite negation in that in both cases, two morphemes conspire to express a single semantic negation. However, in the case of bipartite negation, it is the standard negator that consists of two morphemes. In the case of negative concord, a negative indefinite combines with the standard negator or with another negative indefinite, with only one negation being interpreted semantically.

That combinations of multiple negative indefinites do not contribute multiple semantic negations is shown by the following examples from Italian (25a) and Czech (25b).

- (25) a. **Nessuno** ha visto **niente**.
 no-one has seen nothing
 ‘No-one saw anything.’

- b. **Nikdo** **nic** **ne**-viděl.
 no-one nothing NEG-saw
 'No-one saw anything.'
 'It is not the case that somebody saw something.'

The two negative indefinites in (25a) (marked in bold) together give rise to one single negative meaning. In the Czech example (25b) there are even three negative elements, but only one semantic negation. Of these three elements two are negative concord items, i.e. *nikdo* and *nic*, while *ne* is the standard negator.

Within negative concord, a distinction is made between strict negative concord and non-strict negative concord (Giannakidou, 1997, 2000). In the first case, the sentence negator must always appear when a negative concord item is present. In the case of non-strict negative concord, the sentence negator can be omitted under certain circumstances. Czech is a language with strict negative concord. As we saw above, the sentence negator *ne*- appears as a prefix on the verb (26).

- (26) Pavel ženě peníze **ne**-dal.
 Pavel woman.DAT money.ACC NEG-gave
 'Pavel didn't give money to the woman.'

In combination with negative concord items the negative prefix *ne*- must always appear on the verb, regardless of the position of the negative concord item.

- (27) a. **Nikdo** ženě peníze **ne**-dal.
 nobody woman.DAT money.ACC NEG-gave
 'No-one gave money to the woman.'
 b. Ženě **ne**-dal **nic**.
 woman.DAT NEG-gave nothing
 'He didn't give the woman anything.'

Italian is an example of a language with non-strict negative concord. The sentence negator is *non*, which appears preverbally, (28):

- (28) a. Gianni **non** ha telefonato.
 Gianni not has called
 b. **Non** ha telefonato Gianni.
 not has called Gianni
 'Gianni hasn't called.'

A relevant fact about Italian is that it can have postverbal subjects, as in (28b). When such a subject is a negative indefinite like *nessuno* 'nobody', we get negative concord, as in (29a). However, this is not the case when the negative indefinite precedes the verb (29b), in which case the clausal negator *non* is omitted (or we get a double negation reading when *non* is present).

- (29) a. **Non** ha telefonato **nessuno**.
 not has called nobody
 'No-one called.'
 b. **Nessuno** ha telefonato.
 nobody has called

‘No-one called.’

Since the direct object necessarily follows the verb, it always requires negative concord, as we saw above in (24). Because in Italian negative concord items do not systematically require the presence of the sentence negator, Italian is classified as a non-strict negative concord language.

This kind of negative concord also occurs in many substandard varieties of English. In (30), a number of English song titles are given that illustrate the phenomenon.

- (30) a. Ain’t **no** sunshine (when she’s gone) (Bill Withers, 1971).
b. Ain’t **no** mountain high enough (Marvin Gaye and Tammi Terrell, 1966).
c. (I can’t get **no**) satisfaction (Rolling Stones, 1965).
d. Don’t **Nobody** Bring Me **No** Bad News (Mabel King, 1978)
e. Don’t Come Round Here **No** More (Tom Petty and the Heartbreakers, 1985)

Languages can also combine negative concord and bipartite negation, when their standard negator is bipartite. French and West Flemish are examples of bipartite negation languages (as we saw above), and they also feature negative concord. Negative indefinites in French also show bipartite negation, i.e. they require the presence of *ne*, as (31a) shows. But there is no negative concord with *pas* in French: when there is only one negative indefinite in the sentence, as in (31a), then the addition of *pas* leads to a double negation reading (31b). But with two or more negative indefinites, as in (31c), we do see negative concord, i.e. only one of the negative indefinites is interpreted negatively.⁸

- (31) a. **Personne n’est** venu
no-one NEG=is come
‘No-one came’
b. **Personne n’est pas** venu
no-one NEG=is not come
‘No-one didn’t come.’
c. **Personne n’a rien** vu.
no-one NEG=has nothing seen
‘No-one saw anything.’
‘It is not the case that somebody saw something.’

In West Flemish, on the other hand, negative concord already occurs with one negative indefinite, which can occur together with the standard negator *nie* with a negative concord reading (32a) (in addition to the bipartite negation clitic *en*). Negative concord also occurs with two (or more) negative concord items, as in (32b) (Haegeman, 1995, p. 116).

- (32) a. da Valère ier **niemand nie en**-kent.
that Valère here no-one not NEG-knows
‘that Valère knows no-one here.’

⁸den Besten (1986) and van der Wouden (1995) and van der Wouden and Zwarts (1993) use the term ‘negative spread’ for the co-occurrence of multiple negative indefinites without the presence of an additional standard negator (as in (25a)), and ‘negative doubling’ when the standard negator co-occurs with a negative indefinite, giving rise to one $\neg$ (as in (24)). However, all languages with negative spread also have some type of negative doubling (e.g. even though French *pas* cannot take part in negative concord, *ne* can), we believe negative concord is the better term to refer to the phenomenon.

- b. da Valere woarschijnlijk an **niemand niets** **nie en**=zegt
 that Valère probably to nobody nothing not NEG=says
 ‘that Valère probably doesn’t say anything to anyone.’

Note that the last sentence in (32b) contains four negative elements, which nevertheless are interpreted as a single $\neg$.

Negative concord has been discussed extensively in the literature, since it poses serious challenges to the principle of the compositionality of meaning, “which says that the meaning of a complex whole is a function of the meaning of its parts and the way they are put together” (De Swart & Sag, 2002, p. 375). If indeed we were to put the different negative concord items together to compute the meaning of the whole, and each of those introduced a logical negation, then we would end up with double negations in all of the above examples. In the remainder of this section, we discuss five different proposals which have been suggested in the literature to deal with this problem.

One solution, proposed by Zanuttini (1991) and Haegeman and Zanuttini (1996) is to assume that all negative concord items are negative quantifiers, and that an operation, called *factorization*, reinterprets a sequence of quantifiers that each contain a negative operator as a new sequence with only one operator. However, according to May (1989), factorization fails to preserve compositionality, because a part of the semantics of each negative concord item is actually erased.

A second type of approach assumes that negative concord items are not quantifiers, but Negative Polarity Items (henceforth NPIs; see Sect. 3.2 for discussion; e.g. Giannakidou, 2000; Ladusaw, 1992; Laka, 1990, and many others). This solution basically tries to do away with the compositionality problem by reducing the negative concord item to an existentially quantified NP. However, De Swart and Sag (2002) point out a problem for this assumption. Negative concord items, like French *rien*, can appear as fragment answers (33b), but NPIs like *quoi que ce soit* ‘anything’ cannot (33c).

- (33) a. Qu’est-ce que tu as vu?
 What that you have seen
 ‘What did you see?’
 b. **Rien.**
 nothing
 ‘Nothing.’
 c. *Quoi que ce soit.
 what that it is.SBJV
 ‘Anything’

If negative concord items were like any other NPI, then this contrast is unexpected. Moreover, De Swart and Sag (2002, p. 376) observe that a French example like (34) is in fact ambiguous between a negative concord and a double negation reading (though admittedly the concord reading is definitely the preferred one).

- (34) **Personne (n’)a rien fait.**
 No-one NEG=has nothing done
 ‘No one has done nothing.’ (DOUBLE NEGATION)
 ‘No one has done anything.’ (NEGATIVE CONCORD)

This ambiguity is again hard to explain if negative concord items were NPIs.

A variant of this second approach deals with negative concord in terms of intrinsic ambiguity, either configurational/contextual (van der Wouden, 1994, 1997; van der Wouden & Zwarts, 1993) or lexical (Herburger, 2001). Herburger (2001) proposes that the lexicon of languages with negative concord, like Spanish, contains two different lexical items for the same negative concord item, one being a negative quantifier (to account for fragment answers), and the other a negative polarity item. Support for her approach comes from the fact that this type of situation could be regarded as the Stage III of the Jespersen Cycle, where there is obligatory bipartite negation (see the discussion in Sect. 2.1.2 above). The problem with this type of approach is that it requires a multiplication of homophonous lexical entries. The approach of De Clercq (2019, 2024), couched in Nanosyntax (Starke, 2009), does away with that problem and proposes to capture the ambiguity of negative concord items without creating doublets in the lexicon by treating the identity between the negative quantifier and the concord item as a case of syncretism.

A third type of approach is the one by De Swart and Sag (2002), who account for data of this type in terms of a polyadic quantificational approach, based on May (1989) and van Benthem (1989). Double negation arises in their approach due to interpreting a sequence of negative indefinites as the iteration of monadic quantifiers. Concord arises due to interpreting a sequence of negative indefinites via resumption, i.e. quantification is expressed over multiple variables at once. In other words, De Swart and Sag (2002) introduce two types of quantificational mechanisms to capture the difference between negative concord and double negation. A potential problem with this type of approach is that it predicts that each language should have both readings for sequences of negative indefinites, and that is not the case.

A fourth approach to negative concord is the syntactic agreement approach, which adopts minimalist technology, in particular the distinction between interpretable and uninterpretable features (Penka & Zeijlstra, 2010; Zeijlstra, 2004a, 2009). Negative concord items typically appear to be negative, but are not so interpreted. Under this type of approach, this follows from them having an uninterpretable [uNeg] feature. That is, they are syntactically marked for negation, but semantically non-negative. In strict negative concord languages like Czech, the standard negator, like *ne* in (27), is also argued to be endowed with this [uNeg] feature. A silent negative operator with an interpretable [iNeg] feature will license all the negative concord items in its scope, leading to the effect that only one semantic negation is interpreted. For a non-strict negative concord language like Italian or Spanish, Zeijlstra (2004a) argues that the standard negator has [iNeg], allowing it to license [uNeg] items in its scope. However, when a negative indefinite appears in preverbal position it will need to get self-licensed. This self-licensing mechanism is also called for in contexts like fragment answers (33), raising the question whether these so-called semantically non-negative words are not actually semantically negative after all if they need to trigger the appearance of silent/covert negative operators.

We refer the reader to the literature quoted in this section for more details.

2.3 Negative correlative conjunctions

In this section we discuss a type of negator which shows a formal similarity with the negative indefinites in being composed of *n-*, followed by *either* (as in *neither*), or by the disjunctive conjunction *or* (as in *nor*). *Neither* and *nor* typically occur together in the negative correlative *neither ... nor*, also referred to as negative connectives (Van der Auwera et al., 2021).

- (35) **Neither** Alissa, **nor** Marian went to school today.

The meaning of this sentence involves either a conjunction and two negations, or a disjunction and one negation:

- (36) a. $(\neg p \wedge \neg q)$
 b. $\neg(p \vee q)$

We return to the equivalence between conjunction and disjunction below, when discussing De Morgan's Law. In line with (36a), its meaning can be paraphrased as in (37):

- (37) Alissa didn't go to school today, and Marian didn't go to school today.

Formally, *nor* resembles English negative quantifiers in being a combination of *n-* and a second element (*or*) with existential semantics (compare pairs like *ever-never*, *a-no*, *somebody-nobody*, etc.). The semantics of *or* is existential because there exists a logical equivalence between the predicate logical quantifiers $\forall$ and $\exists$ on the one hand and the Boolean connectors $\wedge$ and $\vee$, respectively. Universal quantification over a finite domain $D = \{a_1 \dots a_n\}$ is equivalent to the conjunction $P(a_1) \wedge \dots \wedge P(a_n)$, and the existentially quantified formula $\exists.P(x)$ is equivalent to the disjunction $P(a_1) \vee \dots \vee P(a_n)$. Similarly, the universally negative $\neg\exists.P(x)$ is equivalent with *neither ... nor*, i.e. $\neg(P(a_1) \vee \dots \vee P(a_n))$. For this reason, we shall refer to this reading as the universally negative reading (see Sect. 3.1 below, where the universally negative reading will be discussed in more detail).

The combination *neither ... nor* as in (35) looks very much like a case of negative concord, with only one semantic negation being contributed by two negative morphemes (and assuming a logical representation with a disjunction in the scope of a single negation, as in (36b)).

2.4 Squatitive negation

Squatitive negation involves a set of words, all derived from taboo-items, that shows the interesting behaviour of being used both in the scope of negation (as negative indefinite or NPI, see Sect. 2.2 and 3.2), as well as with full negative meaning on their own (De Clercq, 2011; Horn, 2001a; Postal, 2004).⁹ The list in (38) is a mixture of American and British English squatitive negation items (based on De Clercq, 2011; McCloskey, 1993; Postal, 2004).

- (38) squat, fuck-all, beans, crap, dick, diddley, diddley-poo, diddley-squat, jack, jack-shit, jack-squat, piss-all, poo, shit, shit-all, sod-all, bugger-all, naff-all, crap-all

Postal (2004) also refers to them as vulgar minimizers, because they share properties with minimizers, like *a drop*, *a finger*, which, like other NPIs, always need to occur in the context of negation.

The examples in (39) (from Postal, 2004, p. 159) illustrate the use of squatitive items in the context of negation, yielding a meaning similar to the negative polarity item *anything*. The examples in (40) show that, when negation is absent, the squatitive items can give rise to a negative meaning on their own (De Clercq, 2011; Sarah Haas, p.c.).

- (39) a. Olmstead didn't say **dick** about the new dean.
 b. Olmstead doesn't understand **squat** about topology.

⁹See also Sailor (2017) for *fuck-inversion*, which is used in some varieties of the English spoken in the British Isles, and seems to behave quite differently from squatitive negation as discussed in this section.

- (40) a. Claudia saw **squat**.
 b. **Fuck-all** happened.
 c. It's 3:15 and I've done **fuck-all** but this thing.

De Clercq (2011, p. 15) also discusses the use of British English squatitive elements as determiners, as in (41) with a meaning similar to *no*.¹⁰

- (41) A: Did Claudia buy everything on the shopping list?
 B: She got the whiskey and the cigars, but she bought **fuck-all** books or magazines.

Squatitive negation is definitely not limited to English. In substandard Dutch a definite expression *de ballen* 'the balls' can be used to negate a sentence (42), and like the English squatitive items, the same noun, but without the definite article, and marked by the negator *geen* 'no' can give rise to emphatic verbal negation.

- (42) Ik heb er **de ballen** verstand van.
 I have R-pron the balls brains of.
 'I don't know the first thing about it!'
- (43) Ik heb er **geen ballen** verstand van.
 I have R.pron no balls brains of
 'I don't know the first thing about it!'

Also in German this type of squatitive items has been discussed. Buschbom et al. (2024) refer to *einen/keinen feuchten Kehricht* in (44a) as implicit negators, and to *einen Scheiss* as in (44b) expressive negators.

- (44) a. Das geht dich (k)einen feuchten Kehrischt an.
 This concerns you (no)/a wet dirt PRT
 'This is none of your business.'
- b. Einen Scheiß muss ich!
 a shit must I
 'I don't have to do ANYthing.'

For the first group the authors argue that they are only semantically negative, while for the latter group they argue they are both syntactically and semantically negative.

While we keep a fine-grained comparison with German and Dutch squatitives for future work, we will investigate in the remainder of this chapter through various semantic and syntactic tests whether English squatitive items, and the other "negative" morphemes discussed in this section also truly contain a semantic negation. We will see that not all negative morphemes behave alike with respect to these tests and that the notion of negativity is more complex than the logical operator $\neg$ suggests.

2.5 Other negative words and expressions

In addition to the negative indefinites, negative correlatives, and squatitive negators of the previous sections, there are other types of words and complex expressions, which may potentially express logical negation, or have some intuitively negative feel to them.

¹⁰Special thanks go to William Harwood for help with the data on British English *squat*.

As a first instance, this category includes complex negative expressions like *by no means*, *in no way*, and *anything but*. As the examples show, these may or may not include overt negative morphology.

- (45) a. It is **by no means** certain that we'll finish the project by June.
 b. She had **in no way** intended to offend anybody.
 c. Nasrin is **anything but** kind.

The negator *anything but* typically combines with predicative adjectives and nouns. As we shall see in Sect. 3, these behave very much like the standard negator and negative indefinites with respect to the diagnostics for negation.

A second category includes the adverbs *hardly*, *rarely*, and *seldom*:

- (46) a. Eric can **hardly** remember the accident.
 b. Anna **rarely** drinks coffee.
 c. Kangaroo rats **seldom** have access to drinking water.

Their properties resemble those of the vague scalar quantifier *few*, which meaningwise is close to *not many* discussed above.

- (47) a. **Few** arrows have hit the target.
 b. She's read **few** books.

A third category are the negative adjectives. These may be morphologically derived from positive adjectives by means of a negative prefix (48a), or they may be semantically negative without such a prefix (48b).

- (48) a. My grandfather was unaware of the problem.
 b. It was difficult for him to stay with her.

The presence of a covert negation in the adjectives of (48b) is suggested by the systematic impossibility of prefixing them with *un-*, as observed by Jespersen (1942, p. 466) (also Zimmer, 1964; Horn, 2001b, p. 275; Horn, 2005).

- | | | | | |
|------|----|------------|----|-----------|
| (49) | a. | unhappy | b. | sad |
| | | unwise | | foolish |
| | | unclean | | dirty |
| | | unfriendly | | hostile |
| | | unhealthy | | sick |
| | | unkind | | rude |
| | | untrue | | false |
| | | uneasy | | difficult |

An account of this pattern in terms of a constraint on structurally adjacent negative heads is proposed in De Clercq and Vanden Wyngaerd (2019) (see also Collins, 2018).

Another example of this category is the preposition *without*, which is clearly negative, in the sense that its meaning can be paraphrased as *not with* (and various other types of evidence support this conclusion, as we will see in Sect. 3). At the same time, its scope does not include the whole sentence, as can be seen from looking at (50).

- (50) The children left for school **without** their school bags.

Here the event of leaving for school is not denied: the children did leave for school, they just did so not taking their school bags.

As with the negative indefinites of the previous section, the words and expressions discussed in this section can be decomposed into a negation and another element, as illustrated in (51).

(51)	negative word	=	X	$\neg$	X
	hardly	=	almost	not	
	rarely	=		not	often
	few	=		not	many
	without	=		not	with
	by no means	=		not	by any means
	in no way	=	not	in any way	
	anything but	=		not	
	un- A_{POS}	=		not	A_{POS}
	A_{NEG}	=		not	A_{POS}

The decomposition is intended to show the relatedness of the pairs in the table, a relatedness which is mediated by a semantic negation. At the same time, it must be observed that not all these negations have the same meanings, in that some of them correspond with logical contradiction (where a sentence and its negation can be neither both true, nor be both false), and others with contrariness (where a sentence and its negation cannot be true, but can be false together). Contrary opposition exists between the pairs *rarely-often*, *few-many*, *unhappy-happy*, and *short-tall*. This implies that *short* is not the semantic equivalent of *not tall*, as a person who is neither short nor tall could be truthfully described as being *not tall*, but not as *short*. We set this issue aside here, referring the reader to De Clercq and Vanden Wyngaerd (2019) for more detailed discussion, as well as a proposal for reducing contrary negation to contradictory negation.

As we shall show below, not all of these negations always take sentential scope. Various tests indicate, for example, that the negative prefix *un-* never takes sentential scope. Similarly, as we noted above, in a case like (50) with *without*, the negation does not negate the event, but has scope over the adjunct. We come back to this in Sect. 3.4.

A category of negative words for which a decomposition is not straightforward includes verbs with a negative meaning but with (mostly) no visible negative morphology, like *dissuade*, *doubt*, *refuse*, *deny*, *forget* (see Ladusaw, 1980 for discussion).

- (52)
- a. I **doubt** that he is telling the truth.
 - b. Luke **denied** that the accusations were true.
 - c. Luke **denied** the accusations.
 - d. She **refused** to kiss him.

In (52a), a paraphrase with *not sure* is possible, and *deny that p* seems equivalent with *claim that not p*, but it is unclear how that paraphrase extends to (52c). For (52d) (*refuse to VP*), the paraphrase with an overt negation could be *not agree to VP*. Observe, however, that such paraphrases work in both directions: if *doubt that p* can be rephrased in terms of *not be sure that p*, then *be sure that p* can be rephrased in terms of *not doubt that p*. The evidence for the presence of semantic negation in the verbs of (52) is therefore not conclusive. Yet there is clearly an intuitive sense in which the verbs of (52) are negative, while their counterparts

in the paraphrases with an overt negation are not. In Sect. 3 we shall discuss a number of diagnostics that put the verbs of (52) firmly in the camp of negation.

3 Diagnosing negation

In the previous section we discussed different types of morphemes, words, and complex expressions, which are all negative in some way. In this section we will discuss a number of tests that provide support to them being classified as negative, while at the same time pointing to subtle differences in what it means to be negative. By this we mean that the diagnostics will show that there are different degrees of negativity, which is a property of negation that the single logical operator $\neg$ is ill-equipped to represent. The tests to be discussed involve the validity of De Morgan’s first law (Sect. 3.1), the licensing of Negative Polarity Items (Sect. 3.2), the interaction with modality (Sect. 3.3), and then also a set of “syntactic” tests (Sect. 3.4)

3.1 De Morgan’s first law

3.1.1 Standard negators

The first test that we want to discuss involves De Morgan’s First Law, which posits a logical equivalence between the Boolean operators $\wedge$ and $\vee$, when they combine with negation:

$$(53) \quad \text{De Morgan’s First Law} \\ (\neg p \wedge \neg q) \Leftrightarrow \neg(p \vee q)$$

Named after the 19th century British mathematician Augustus De Morgan, it forms a pair with his second law in (54).

$$(54) \quad \text{De Morgan’s Second Law} \\ (\neg p \vee \neg q) \Leftrightarrow \neg(p \wedge q)$$

The Boolean operators $\wedge$ and $\vee$ are said to be each other’s dual, i.e. one can always be translated into the other using logical negation, as shown in (55):

$$(55) \quad \begin{array}{ll} \text{a.} & (p \vee q) \Leftrightarrow \neg(\neg p \wedge \neg q) \\ \text{b.} & (p \wedge q) \Leftrightarrow \neg(\neg p \vee \neg q) \end{array}$$

The same relationship of duality holds between the universal and existential quantifiers:

$$(56) \quad \begin{array}{ll} \text{a.} & \forall x.P(x) \Leftrightarrow \neg\exists x.\neg P(x) \\ \text{b.} & \exists x.P(x) \Leftrightarrow \neg\forall x.\neg P(x) \end{array}$$

Finally, as we shall see in Sect. 3.3, a dual relationship also exists between necessity and possibility modals (see Demey and Smessaert, 2016 for discussion of more duality phenomena in logic and natural language).

We will be concerned in particular here with De Morgan’s First Law. This equivalence between *and* and *or* in the presence of a negation can also be observed in natural language. One way to formulate (53) is that an *and* that has two negations in its scope turns into an *or* when the negation takes scope over the logical connective. We start out by looking at simple sentences with the standard negator, which we assume to undoubtedly contain a $\neg$. In (57) we see two NPs conjoined by the coordinating conjunction *or*.

(57) Nasrin doesn't like [beer **or** wine].

Although (57) contains *or*, its meaning is more like that of *and*, in the sense that the sentence is true if neither of the conjuncts is true, i.e. (57) means that Nasrin does not like beer, and he does not like wine. We also see this *and* appear in (58), which has the same meaning as (57).

(58) [Nasrin doesn't like beer] **and** [Nasrin doesn't like wine].

In (58) we see two negations, one in each conjunct, the two conjuncts being connected with the coordinating conjunction *and*. Formulated in terms of scope, the conjunction *and* (logical $\wedge$) in (58) has scope over the negations, as in the statement that is to the left of the material equivalence symbol $\Leftrightarrow$ in (53) above.¹¹ The sentence (57), with *or*, in contrast, has the disjunctive operator in the scope of the negation (as in the part that is to the right of the material equivalence arrow in (53) above). The latter situation always involves some kind of ellipsis (understood in its broadest sense), as we can see when comparing (58) and (57).

Having established that the appearance of *or* (logical $\vee$) means something that semantically corresponds to *and* in the presence of negation and that this can be understood in terms of the Morgan's First Law, we can also use this property as a diagnostic to detect the presence of a (semantic) negation, or the $\neg$ from classical logic, in those places where there is no standard negator but some alternative negative word, like a negative indefinite, adverb, or whatever type of intuitively negative element. When *or* has a meaning that can be paraphrased with *and*, this points to the presence of an underlying negation.¹² Let us therefore look at the negative words that we discussed in Sect. 2 through the lens of this diagnostic.

3.1.2 Negative indefinites

We start with negative quantifiers in English. The same thing we saw with the standard negator applies to negative indefinites, such as *no-one*. Here too, (59a) has the same meaning as (59b).

- (59) a. **No-one** likes Bach **or** Wagner.
 b. **No-one** likes Bach **and** **no-one** likes Wagner.

From this we can conclude that negative indefinites contain a logical negation, and that in (59a) this negation has scope over the conjunction *or*. This is confirmed by the fact that, if we replace the *or* in (59a) with an *and*, as in (60), then the meaning changes: nobody likes both, i.e. both Bach and Wagner.

(60) **No-one** likes Bach **and** Wagner.

For example, in a family where everyone likes Bach but hates Wagner, (59) is false, but (60) is true.

As is to be expected, a sentence with *or* but without a negative quantifier behaves very differently from one with a negative quantifier: the sentences in (61) do not have the same meaning.

(61) a. **Everyone** likes Bach **or** Wagner.

¹¹The notation in (53) does not show the scope of logical $\wedge$ visually, but in the Polish/Łukasiewicz notation the scope of $\wedge$ (symbol K for conjunction) over the negations (symbol N) is more clearly visible: KNpNq.

¹²This observation is also made in Reiss (to appear).

- b. **Everyone** likes Bach **and** **everyone** likes Wagner.

Consider again our family where everyone likes Bach but hates Wagner: in such a situation, (61a) is true, but (61b) is false. The reason for this is of course that the universal quantifier does not have a hidden negation inside it, and so no change from *or* to *and* or vice versa can occur, because De Morgan's First Law does not apply.

The other negative indefinites of (20) behave like *no-one*, i.e. they give rise to a universally negative meaning when there is an *or* in their scope. This is illustrated in (62):

- (62) a. **Nothing** could interest the boys **or** the girls.
 b. He bought **no** books **or** magazines.
 c. I **never** drink beer **or** wine with my dinner.
 d. **Nowhere** in this town can you buy beer **or** wine.
 e. **No** student **or** professor knew the answer.

For example, sentence (62a) means that there was nothing to interest the boys, and there was nothing to interest the girls (i.e. the universally negative meaning), and so do the other sentences in (62). These sentences contrast minimally with those in (63), where the negative indefinite is replaced by a non-negative counterpart:

- (63) a. **Many things** could interest the boys **or** the girls.
 b. He bought **some** books **or** magazines.
 c. I **always** drink beer **or** wine with my dinner.
 d. You can buy beer **or** wine **everywhere** in this town.
 e. A student **or** a professor knew the answer.

Here we have a reading in which it is not necessary that there are many things that could interest the boys, and many that could interest the girls. It is sufficient that the sums of their interests can be characterized as 'many things' in some way. This is most clearly the case in (63c), where it is a very natural interpretation that you always drink one or the other (beer or wine) with your meal, but not both.

3.1.3 Squatitive negation

Next, we look at the squatitive negators, which we introduced in Sect. 2.4. These elements behave in exactly the same way with respect to the *De Morgan* test as the standard negator and the negative indefinites, like *no*, *nothing*, *without*, etc.

- (64) **Fuck-all** students smoke **or** drink.
 ↔ **Fuck-all** students smoke, **and** **fuck-all** students drink.
- (65) Claudia bought **fuck-all** books **or** magazines.
 ↔ Claudia bought **fuck-all** books and Claudia bought **fuck-all** magazines.

That is, the equivalence between the sentences with *and* and with *or* holds in all contexts. We conclude that they are fully negative.

3.1.4 Other negative words and expressions

We finally turn to the category of the other negative words and expressions which we discussed in Sect. 2.5 above. These fall into two groups. In a first group, De Morgan's first Law

applies in full. However, there is also a group where the logical equivalence between *and* and *or* does not hold, suggesting that they are either nonnegative, or, alternatively, negative in a different manner.

We start off the discussion with the first group, where De Morgan holds in full. A case in point are complex expressions of the type *anything but*, *in no way* and *by no means*. We show this for *anything but* in (66), leaving it to the reader to reproduce it for the other two members of the triplet.

- (66) a. Nasrin is **anything but** kind **or** gentle.
 b. Nasrin is **anything but** kind **and** Nasrin is **anything but** gentle.

The preposition *without* also shows full De Morgan behaviour, as shown in (67).

- (67) The children went to school **without** a coat **or** bag.
 ↔ The children went to school **without** a coat, **and** the children went to school **without** a bag.

The same is true for all sorts of negative adjectives, whether prefixed or not, as long as they can be complemented by an infinitival clause. The negation inside the adjective seems to be able to turn the *or* into an *and*, yielding a universally negative meaning.

- (68) It is **impossible** to find beer **or** wine here.
 ↔ It is **impossible** to find beer here, **and** it is **impossible** to find wine here.
- (69) It is **difficult** to find beer **or** wine here.
 ↔ It is **difficult** to find beer here, **and** it is difficult to find wine here.

With negative verbs like *refuse* and *deny* De Morgan's First Law also holds:

- (70) Max **refuses** to drink **or** smoke.
 ↔ Max **refuses** to drink **and** Max **refuses** to smoke.
- (71) The postman **denies** that he smokes **or** drinks.
 ↔ The postman **denies** that he smokes **and** the postman **denies** that he drinks.

We now turn to the second group of negative words, which behave differently, in the sense that the sentence with one negation and *or* does not mean the same as the sentence with two negations and *and*. In other words, De Morgan's equivalence does not always hold. An example of this is the vague quantifier *few* (see also Solt (2006)):

- (72) **Few** students smoke **or** drink.
 ↗ **Few** students smoke, **and** **few** students drink.

If the first sentence is true, then the second is always true, but the reverse does not hold. There are cases in which the second sentence (with *and*) is true, but the one with *or* is not. Suppose we have a group of seven students, two of whom smoke (and do not drink), two others drink (and do not smoke), and the remaining three neither smoke nor drink. A majority of four out of seven therefore smokes or drinks. In such a situation, the sentence with *and* is clearly true (there are few smokers and few drinkers), but the sentence with *or* is not (since there is a majority that either smokes or drinks). It seems that the *or* in the scope of *few* is a real semantic *or*: the sum of smokers and the drinkers has to amount to a number that can be

considered ‘few’ for the first sentence of (72) to be true, and to count that sum, we count the students who smoke or drink, not those who smoke and drink.¹³ In contrast, for the sentence with *and* to be true, the number of smokers has to qualify as ‘few’, and the number of drinkers has to qualify as ‘few’ as well, but we do not look at the total number of students who smoke or drink. In requiring reference to the sum of the smokers and the drinkers, *few* resembles *many*:

(73) **Many** students smoke **or** drink.

If the sum of the smokers and the drinkers amounts to four out of seven (as in the situation just sketched), we would be likely to say that (73) is true (whereas (72) with *or* is false). Note finally that (72) with *or* behaves like (74) with respect to De Morgan:

(74) **Not many** students smoke **or** drink.

The relevance of this observation is that (74) clearly contains an overt negator, yet it fails the De Morgan test, like *few*, which does not contain an overt negator. Rather, it looks like there are different degrees of negativity, which are in part related to the type of negator, and in part to its syntactic position. This conclusion will be further confirmed by our discussion of negative polarity in Sect. 3.2 below.

The adverbs *hardly*, *rarely*, and *seldom* also fit into the category of non-De Morgan compliant negators. We first show this for (75):

(75) You can **hardly** find beer **or** wine here.
 ↗ You can **hardly** find beer here, **and** you can **hardly** find wine here.

Here too, it is true that the truth of the first sentence implies the truth of the second. But the reverse is not so clearly the case: if you can find beer in 2 out of 7 shops in a city, and wine in 2 others, then you can find either beer or wine in 4 out of 7 shops. In that case, the second sentence is true, but the first untrue. Similar contexts can be created for *rarely*:

(76) I **rarely** drink beer **or** wine with my dinner.
 ↗ I **rarely** drink beer with my dinner **and** I **rarely** drink wine with my dinner.

Here too, there are conceivable circumstances in which the second sentence is true, and the first is false. Sketching such a context requires certain assumptions about where the threshold lies for speaking of ‘rarely’. Suppose that drinking beer (or wine) with your meal three times a month counts as ‘rarely’, but not six times. If someone then drinks beer with his meal three times a month, and also drinks wine three times a month, then the second sentence is true, but the first is not.

In the Sect. 3.2, we will come back to quantifiers like *few*, *hardly* and *seldom*, and how they differ from the ones for which De Morgan’s equivalence does hold. We will call the latter class anti-additive (AA) (Zwarts, 1986), and the former downward entailing (DE) (Ladusaw, 1980). These two classes also have different properties when it comes to licensing NPIs.

¹³The quantifiers *few* and *many* are like gradable adjectives in that they refer to an implicit contextual norm or standard or comparison class for their interpretation (Klein, 1980; Wheeler, 1972). In the text, we are operating in a context where a proportion of two out of seven makes the sentence with *few* true, while four out of seven makes it false. Since these proportions are not an absolute property of this quantifier, the truth conditions may vary with the contextual standard.

3.1.5 Interim summary and outlook

We have found that, for the vast majority of negative expressions, it is the case that *or* has a universally negative reading, i.e. it is subject to De Morgan's First Law, making it logically equivalent to a paraphrase with *and*. But there is also a small group of negative expressions, which we will refer to as the downward entailing (DE) QPs, in which De Morgan's equivalence does not hold true in all situations. In the next section we take a closer look at the phenomenon of Negative Polarity, as this serves as a second diagnostic to detect the presence of a semantic negation. That section will also further develop the concept of negations of different strengths.

3.2 NPI-licensing

In our previous discussion, we have already mentioned Negative Polarity Items (NPIs) a number of times. These are expressions that can only occur in certain kinds of environments (see Buysens, 1959; Giannakidou, 1997; Homer, 2021; Ladusaw, 1980; van der Wouden, 1994, for discussion). Many of these environments can be characterized as 'negative' in a way that is intuitively obvious. For this reason, these items are called Negative Polarity Items: they require a negative pole in the sentence.¹⁴ In the examples below, the NPIs are italicized, and the negative elements bolded (the examples in this section have been inspired by Homer, 2021; Ladusaw, 1980; van der Wouden, 1994).

- (77) a. Sally didn't eat *any* breakfast yesterday.
b. *Sally ate *any* breakfast yesterday.
- (78) a. Joe isn't *all that smart*.
b. *Joe is *all that smart*.
- (79) a. I **never** want to talk to him *anymore*.
b. *I want to talk to him *anymore*.
- (80) a. The children **cannot** *stand* the schoolmaster.
b. *The children *can stand* the schoolmaster.
- (81) a. You **needn't** be afraid.
b. *You *need* be afraid.
- (82) a. **No-one** told the secret to *anyone*.
b. *Everyone told the secret to *anyone*.
- (83) a. **Nobody** *lifted a finger* to help her.
b. *The children *lifted a finger* to help her.

A comparison of the a- and b-sentences shows that the bolded negative word, which is either a standard negator or a negative indefinite, is required for the NPI to be allowed, i.e. the NPI cannot appear when there is no negative word present. NPIs can be expressions like *any*, or collocations such as *can stand* or *lift a finger*. The modal verb *need* (81) is also an NPI when it is used as a modal, i.e. without *do*-support and with a bare infinitive (Iatridou & Zeijlstra,

¹⁴The NPI also needs to be c-commanded by its licenser in many languages (see Homer, 2021; Ladusaw, 1980). We will not discuss this issue further in this paper.

2010).¹⁵ It is the negative counterpart of the obligation or necessity modal *must*, which we formally represent as $\Box$ (see Sect. 3.3 for more discussion). The negation in (81) takes scope over the necessity modal, resulting in a meaning of absence of necessity or obligation.

- (84) Anna needn't worry.
 $\neg\Box W(a)$

The fact that NPIs require a semantic negation to be licensed makes them a useful test to diagnose the presence of $\neg$. The application of this test will reveal, however, that not all NPIs behave in the same way. Observe the difference between (85a) and (85b).

- (85) a. **None** of the girls *give a damn* about the way they look.
 b. *?Few girls *give a damn* about the way they look.

We see that the NPI *give a damn* can occur with a negative indefinite like *none* (85a), but not with *few* (85b).¹⁶ In contrast, an NPI such as *can stand* can occur with both negative indefinites and with *few*, as we see in (80a) and (86b), respectively.

- (86) a. **No** one *can stand* the schoolmaster.
 b. **Few** children *can stand* the schoolmaster.

The property that sets apart the negative indefinite and a negative quantity word like *few* is called **anti-additivity** (Zwarts, 1986). Standard negators and negative indefinites are anti-additive (AA), whereas quantifiers like *few* are not.¹⁷ The property of anti-additivity is in fact one that we discussed extensively in Sect. 3.1 above on De Morgan's First Law. Anti-additive negators are those for which De Morgan's Law holds, i.e. which trigger a universally negative reading for *or*. These negators can combine with any kind of NPI, while negators that are not anti-additive can only occur with a subset of the NPIs. This property distinguishes weak NPIs (which can occur under any negator) from strong NPIs (which only occur under anti-additive negators). The next question is what both groups of negators have in common. This question is relevant, since affirmative sentences do not allow any NPIs at all, as we saw in the above examples (77)-(83). That implies that all types of negative sentences share a property that they do not share with affirmative sentences.

The property in question is called **downward entailment** (DE), and it is opposed to upward entailment (UE), which is a property of affirmative environments. As the name suggests, downward and upward entailment have to do with the kind of inference patterns that sentences give rise to. These different types of inference patterns can be illustrated by the sentences in (87).

- (87) p All children in the class are wearing a woollen sweater.
 q All children in the class are wearing a sweater.

If the sentence p is true, then q must also be true. The converse is not the case: if q is true,

¹⁵When followed by a *to*-infinitive, *need* requires *do*-support, and is not negatively polar, e.g. *You (don't) need to visit her soon*.

¹⁶But see Sedivy (1990) and Sailer (2021, 2022) for a discussion of the use of *give a damn*-type NPIs that can get licensed by 'a conventionalized side message that has the status of a conventional implicature or use-conditional meaning' (Sailer 2022, p. 735).

¹⁷Actually standard negators like *not* are antimorphic, and they are able to license superstrong NPIs like *a bit*. We abstract away from discussing this subtype of anti-additivity in this paper.

then p can be either true or false. This is a typical case of logical entailment: $p \rightarrow q$.

The interesting fact is now that the direction of entailment is reversed when we replace the universal quantifier *all* by the universally negative *no*.

- (88) p No child in the class is wearing a woollen sweater.
 q No child in the class is wearing a sweater.

In this case, we can conclude from the truth of q that p is true. In other words, the negative indefinite *no* reverses the direction of inference: $q \rightarrow p$.

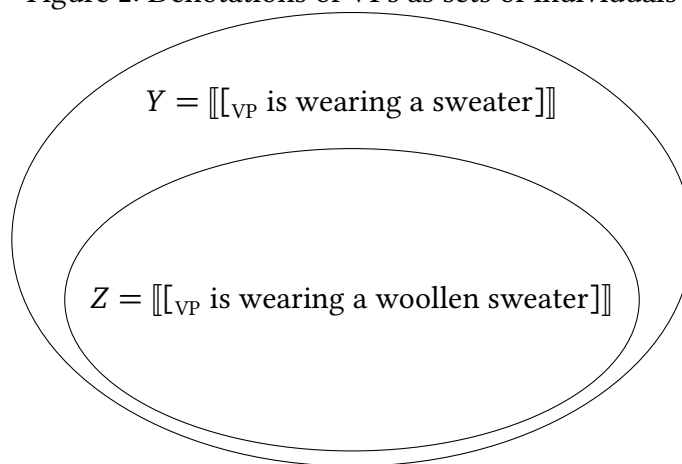
This reversal of the direction of entailment is something that is characteristic of all sorts of negative environments. We can demonstrate this, for example, with *few*, which we saw earlier can license weak NPIs. In (89) we see that the inference pattern is $q \rightarrow p$.

- (89) p Few children in the class are wearing a woollen sweater.
 q Few children in the class are wearing a sweater.

Negative contexts (and others in which the direction of inference is reversed) are downward entailing, or monotonically decreasing. This means that they give rise to inferences from sets to subsets. They are opposed to contexts that are upward entailing (monotonically increasing), such as the affirmative sentences in (87), where the inferences go from the subset to the set.

To understand this, we need to consider VP denotations. VPs in formal semantics can be viewed as sets of individuals, for example the set of the individuals who wear a (woollen) sweater in the above examples. Now the set of individuals who wear a woollen sweater is a subset of the set of individuals who wear a sweater, i.e. the denotations of the VPs in (88) (and therefore also (89)) stand to each other in a set-subset relation. This is shown in the form of a Venn diagram in Figure 2.

Figure 2: Denotations of VPs as sets of individuals



With this in mind, we can better understand the concept of upward and downward entailment: when inferences run from set to subset (i.e. from Y to Z), we are talking about downward entailment. In the case of upward entailment, the reverse is true: inferences run from subset to set (i.e. from Z to Y).

The two sets in Fig. 2 are far from the only ones: within the wearers of a woollen sweater, we can distinguish a smaller subset of the wearers of a green woollen sweater, or of a green

woollen sweater with a V-neck, etc. Conversely, the sweater-wearers are a subset of the wearers of a garment. We can list sentences with all these VPs, and we see that indeed for *all* (and universal quantifiers in general) there is upward entailment: if the first sentence (90a) is true, then all other sentences of (90) are also true.

- (90) a. All children in the class are wearing a green woollen sweater.
 b. All children in the class are wearing a woollen sweater.
 c. All children in the class are wearing a sweater.
 d. All children in the class are wearing a garment.

As before, if we make (90) negative by replacing *all* with *no*, the direction of entailment is reversed, from (90d) to the other sentences.

Anti-additivity is a special case of downward entailment, in the sense that all anti-additive contexts are downward entailing, but not the other way round: not all downward entailing contexts are also anti-additive.

Returning now to the distribution of NPIs, Table 1 shows how their distribution is on the one hand restricted by the entailment patterns triggered by the licenser (downward entailing or DE vs. upward entailing or UE), and on the other hand by the exact nature of the licenser, i.e. whether the licenser is anti-additive (AA), triggering a universally negative reading for *or*, or just DE; note that the column heading ‘DE’ should be understood to mean ‘DE but not AA’, as AA implies DE.

Table 1: Weak and strong NPIs

	AA	DE	UE
strong NPI	✓	✗	✗
weak NPI	✓	✓	✗

Formulated as a generalisation, we can formulate the findings of Table 1 as in (91):

- (91) a. A weak NPI occurs in a downward entailing environment.
 b. A strong NPI occurs in an anti-additive environment.

In (92a-b) we give an overview of different types of negative expressions, classified according to the criterion of whether they are anti-additive, simply downward entailing, or upward entailing.

- (92) a. anti-additive (AA)
not, no, never, nowhere, neither ... nor, by no means, without, refuse, deny, un-A, iN-A, hard, difficult, ...
 b. downward entailing (DE)
few, hardly, seldom, rarely, not everyone, not all N
 c. upward entailing (UE)
some, many, all, everyone, always, often, sometimes

In (93) we sort NPIs according to their strength.

- (93) a. weak NPIs
any, ever, can stand, all that A, at all, need, yet, anymore, ...

- b. strong NPIs
lift a finger, give a damn, either, until (with activity verb), in months, ...

We illustrate some of these combinations in the examples below. Consider a strong NPI such as *until*: it needs a negator that is not only downward entailing, but also anti-additive. The negative element *few* is not anti-additive, as we saw earlier (Sect. 3.1), although *few* is downward entailing. For that reason, (94a) is ruled out (or at least less good than (94b) for most speakers). A weak NPI such as *can stand*, on the other hand, can do with a downward entailing environment, as (94c) shows. For the sake of completeness, (94d) shows that a weak NPI is ruled out in an upward entailing environment.

- (94) a. *?Few guests will arrive *until* Sunday.
- b. Nobody will arrive *until* Sunday.
- c. Few children *can stand* the schoolmaster.
- d. *All children *can stand* the schoolmaster.

Similar contrasts can be reproduced for a number of the other combinations of negators and NPIs.

- (95) Strong NPI: *lift a finger*
 - a. *Not all bystanders *lifted a finger* to help the poor man.
 - b. They looked at the scene **without** *lifting a finger* to help the poor man.
 - c. They **refused** to *lift a finger* to help him.
 - d. *We *lifted a finger* and then resumed our journey.
- (96) Strong NPI: *in months*
 - a. *Not everyone has spoken to him *in months*.
 - b. No-one has spoken to him *in months*.
 - c. Neither Alissa nor Marian have spoken to him *in months*.
 - d. *Many colleagues have spoken to him *in months*.

All these sentences contain a strong NPI. The (a)-sentences contain a licenser or negator that is downward entailing (but not anti-additive); because the NPI is strong, this results in an ungrammatical outcome. The (b-c)-sentences replace the licenser with an anti-additive one, so that the sentence becomes good. The (d)-sentences are upward entailing with an NPI, which is also ungrammatical.

In (97) we have a weak NPI, which combines with all types of downward entailing contexts, whether anti-additive or not (97a-b), but not with upward entailing environments (97c).

- (97) Weak NPI: *ever*
 - a. No-one has *ever* properly looked at this problem.
 - b. He's *barely ever* at home.
 - c. *I want to *ever* visit Moscow.

Note that negative adjectives without any overt negative morphology, like *hard* or *annoyed*, can also license NPIs, as illustrated in (98) (from Ladusaw, 1980, p. 102), with weak NPIs) and (99) (with strong NPIs). In other words, they behave as predicted given that they are anti-additive elements.

- (98) a. It is **hard** to find *any* squid at Safeway *anymore*.
 b. He is **annoyed** that *anyone* would be *all that* anxious to leave.
- (99) a. Due to the war, they were **unable** to speak to him *in months*.
 b. It was **hard** for them to *lift a finger* to help her.

There is one group of negative words, the squative negators, that do not seem to be consistent licensors of NPIs. For Postal (2004, p. 166), *squat* cannot license weak nor strong NPIs (100), but the British English *fuck-all* can license both weak and strong NPIs, both as a determiner and as a noun, as illustrated in (101) (p.c. William Harwood).

- (100) a. *Helga gave **squat** to *anyone*.
 b. *Helga has said **squat** to me *in years*.
 c. *Helga learned **squat** in *any* convent *at all*.
- (101) a. We were all expecting Helga to pass the bottle round but in the end she gave **fuck-all** whiskey to *anyone*.
 b. Helga has said **fuck-all** to me *in years*.

Yet it is clear that squative negators give rise to downward entailment, as in the following examples, where $p \rightarrow q$.

- (102) p Claudia bought **fuck-all** books.
 q Claudia bought **fuck-all** novels.

This suggests that the condition of downward entailment may not be a sufficient condition for NPI-licensing. A possible explanation for the fact that *squat* cannot license NPIs may lie in the fact that squative themselves also frequently (and possibly even predominantly) appear as NPIs, as we pointed out in Sect. 2.4 above. While more data and research are needed to say anything firm about this, it is likely that *squat*, as in (100), has not fully developed into a real negator yet, while *fuck-all* has. The functional instability of *squat* could be the reason that it cannot license NPIs.

As a final point in this section, we need to point out that the condition of being downward entailing is not a necessary condition for NPIs either. A case in point are polarity questions, which can license weak NPIs (103), though not strong ones (104). For direction of entailment to be established, a sentence needs a truth-value. Since questions are neither true nor false, a direction of entailment cannot be determined for them.

- (103) a. Do you know *anything* about the licensing conditions of NPIs?
 b. Is she *all that* clever, you think?
 c. Don't they have *any* shame *at all*?
- (104) a. *Will he arrive *until* Sunday?
 b. *Did she talk to her mother *in weeks*?
 c. *Should they come *either*?

Because of facts like these, Giannakidou (1998) refers to the type of NPIs that can occur in environments that are not DE as PIs, i.e. as polarity items, identifying the licensing property as non-veridicality. Since the complexity of the issue goes far beyond the limits of this chapter, we refer the reader to the literature already mentioned, and in particular to Gajew-

ski (2011), Giannakidou (2011), Homer (2021), and Penka (2020) for overviews of the existing literature on NPIs and PIs. We also refer the reader to the literature on exhaustivity for more background on how an exhaustivity operator can be used to account for NPI distribution, more in particular to capture those cases in which NPIs are blocked in DE or non-veridical environments (Chierchia, 2006; Gajewski, 2011; Guerzoni & Sharvit, 2007; Krifka et al., 1995).

Summarising this section, we saw that NPIs may be used to diagnose negative environments. The distribution of NPIs reinforced the need for a distinction between the different types of negations that we also saw on the basis of De Morgan’s Law: those which are merely downward entailing, and those which are also anti-additive. In the next section we look at yet another diagnostic for negation: modality.

3.3 Interaction with modality

Negation interacts semantically with modality (Butler, 2003; Cormack & Smith, 2002; De Haan, 1997; Francis & Iatridou, 2020; Horn, 2001b; Iatridou & Sichel, 2011; Iatridou & Zeijlstra, 2013; Lyons, 1977, and many others), and this interaction, when observed, can also provide a diagnostic as to the presence and scope of negation in sentences. We will look at the interaction of negation with possibility and necessity modality, two central notions in traditional modal logic. Just like conjunction and disjunction are logically related by negation (see the discussion of duality in Sect. 3.1 above), so are necessity and possibility (Lyons, 1977, p. 787). The relevant logical equivalencies are shown in (105) (necessity being indicated with $\Box$ and possibility with $\Diamond$):

- (105) a. $\Box p \Leftrightarrow \neg \Diamond \neg p$
b. $\Diamond p \Leftrightarrow \neg \Box \neg p$

These in turn translate into (106):

- (106) a. $\Box \neg p \Leftrightarrow \neg \Diamond p$
b. $\Diamond \neg p \Leftrightarrow \neg \Box p$

The first of these two equivalencies (106a) represents an impossibility: the necessity that not p , or, equivalently, the impossibility that p . The second (106b) represents an absence of necessity: it is not necessary that p , or it is possible that not p . Accordingly, both types of meaning can be expressed using either a necessity or a possibility modal, as we shall show below. We will primarily focus on the first of these two equivalencies to diagnose the presence of negation, while noting that the second can serve the same purpose.

Let us move away from the rather abstract depiction in (106), and have a more concrete look at how modal verbs interact with negation in English. Taking the possibility modal *can/could* as an example, we see that it can have a deontic meaning corresponding to permission, or an epistemic reading corresponding to possibility, as illustrated in (107).¹⁸

- (107) $\Diamond p$
a. Anna can borrow my car.
‘It is permitted that Anna borrows my car.’

¹⁸To keep the discussion manageable, we ignore here dynamic modality readings, which refer to an ability, as in *Johnny can swim 100m in less than a minute*. Also note that the present tense modal *can* does not have an epistemic sense in affirmative sentences, but only in questions and negative sentences, i.e. it is an NPI.

- b. Anna could be working from home today.
'It is possible that Anna is working from home today.'

When a sentence containing *can* or *could* is combined with negation, as in (108), negation typically takes semantic scope over the modal, resulting in an impossibility reading.

- (108) $\neg\Diamond p$ ($= \Box\neg p$)
- a. Anna cannot borrow my car.
'It is not permitted that Anna borrows my car.'
 - b. Anna cannot be working from home today.
'It is not possible that Anna is working from home today.'

The same meaning can be conveyed with a necessity modal, due to the fact that now the negation takes scope under the modal:

- (109) $\Box\neg p$ ($= \neg\Diamond p$)
- a. Anna mustn't borrow my car.
'Anna is obliged not to borrow my car.'
 - b. That restaurant must not be any good, it's always empty.¹⁹
'It is necessarily the case that this restaurant is not any good.'

Observe that the English necessity modals *must*, *should*, and *ought* cannot take scope under a clausemate negation. To express absence of necessity, the modal *need* is resorted to instead (see Palmer, 2001 for discussion).

- (110) $\neg\Box p$
- a. Anna doesn't need to borrow the car.
'Anna is not obliged to borrow the car.'
 - b. Anna doesn't need to be working from home today.
'It is not necessary that Anna works from home today.'

Iatridou and Zeijlstra (2010) argue that *must* is a Positive Polarity Item (PPI), which always needs to escape the scope of negation, and hence cannot give rise to absence of necessity readings. Not all languages function in this way: in Flemish Dutch the necessity modal verb *moeten* 'must' only takes scope under negation. The expression of a prohibition requires the possibility modal verb *mogen* 'may'.

- (111) a. $\Box\neg p$
Oscar moet vandaag niet naar school gaan.
Oscar must today not to school go
'Oscar doesn't need to go to school today.'
- b. $\neg\Diamond p$ ($= \Box\neg p$)
Oscar mag vandaag niet naar school gaan.
Oscar may today not to school go
'Oscar can't go to school today.'

¹⁹According to Swan (2005, p. 337) (from which the example in (109b) is taken), the epistemic impossibility reading of *must not* is restricted to American English. Even there, the more common way to express epistemic impossibility would be with *cannot*.

The same pattern holds in German (Iatridou & Zeijlstra, 2013, p. 530).

Also note that the impossibility reading does not always arise when a possibility modal is negated. This is notably the case with *may/might*, which takes scope over negation in its epistemic sense (112b), but below negation when it has a deontic meaning (112a):

- (112) a. $\neg\Diamond p$ ($= \Box\neg p$)
 Anna may not borrow my car.
 ‘It is not permitted that Anna borrows my car.’
 b. $\Diamond\neg p$
 Anna may not be working from home today.
 ‘It is possible that Anna is not working from home today.’

The possibility modal *can* is in fact ambiguous, and can scope both under a negation (113a), or over it (113b) (examples from Horn, 1972, 1975).

- (113) a. $\neg\Diamond p$
 A good Christian **cannot/can’t** attend church and still be saved.
 b. $\Diamond\neg p$
 A good Christian can **not** attend church and still be saved.

The example (113a) represents the most common situation for the scope of the modal, where the negation scopes over the modal. It has the unlikely reading that it is impossible for a good Christian to attend church and still be saved. In (113b), the modal takes scope over the negation, yielding the more likely reading, where it is possible for a good Christian not to attend church and still be saved. Note that in this case, the negator cannot be enclitic, so that these examples are not true minimal pairs.

Table 2 gives an overview of the various types of modal meanings, their interaction with negation, and which English modals can express which types of meaning. Cells in the table that show a mix of possibility and necessity modals are bolded.

Table 2: Types of modal meaning in interaction with negation		
	EPISTEMIC	DEONTIC
$\Box p$	necessity (<i>must, should, need, have to, ought to</i>)	obligation (<i>must, should, need, have to, ought to</i>)
$\Diamond p$	possibility (<i>could, may, might</i>)	permission (<i>can, could, may</i>)
$\neg\Box p \equiv \Diamond\neg p$	absence of necessity (<i>don’t need to, may not</i>)	absence of obligation (<i>don’t need to, don’t have to</i>)
$\neg\Diamond p \equiv \Box\neg p$	impossibility (<i>cannot, couldn’t, must not^{AE}</i>)	prohibition (<i>cannot, couldn’t, shouldn’t, mustn’t, may not</i>)

In what follows we show how the patterns of interaction between modality and standard negation that are summarised in Table 2 are reproduced with the other negative words that

we discussed before. We start by zooming in on negative indefinites, and the negative correlative pair *neither ... nor*. Just like with the standard negator, the possibility modal *can/could* typically scopes under negation, giving rise to prohibition (deontic) or impossibility (epistemic) readings, as illustrated in (114):

- (114) $\neg\Diamond p$
- a. **No-one** can leave the building.
 - b. The journalists could take **nothing** into the briefing.
 - c. Charles can **never** drink alcohol.
 - d. You can park **nowhere** on the campus.
 - e. **No residents** can still be in the building.
 - f. **Neither** Alissa, **nor** Marian can be at school today.

In contrast, the necessity modals like *must*, *should* and *ought to* take scope over negation, giving rise to readings that are logically equivalent to those of (114).

- (115) $\Box\neg p$
- a. **No-one** must leave the building.
 - b. The journalists should take **nothing** into the briefing.
 - c. Charles must **never** drink alcohol.
 - d. You ought to park **nowhere** on the campus.
 - e. **No residents** should still be in the building.
 - f. **Neither** Alissa, **nor** Marian should go to school today.

The examples in (114) and (115) contrast semantically with those in (116), which instead of negative indefinites contain other types of quantificational elements, which are not negative, and which give rise to permission or possibility readings:

- (116)
- a. **Everyone** can leave the building.
 - b. The journalists could take **a few items** into the briefing.
 - c. Charles can **always** drink alcohol.
 - d. You can park **everywhere** on the campus.
 - e. **Most residents** could still be in the building.

For at least some of the negative indefinites in (115) we can create contexts in which a scope ambiguity arises.²⁰ This is the case with (117a), where the part between brackets disambiguates the sentence in the direction of the permission/possibility reading, i.e. the one where the modal has scope over the negation.

- (117)
- a. In certain cases a doctor can do **nothing** (and that may even be the recommended course of action).
 - b. Fred could invite **no-one** to his birthday party (and celebrate it all by himself).

A number of the other anti-additive expressions and downward entailing words and expressions that we discussed above also give rise to prohibition or impossibility readings in

²⁰Note that the ambiguity tends to arise mostly when the negative indefinite is in an object position. Subject negative indefinites outscope *can*, and only scope under deontic modals like *must*, *should* and *ought to* and some raising predicates. Moreover, Subject NegDPs never fully reconstruct, only the indefinite part reconstructs giving rise to split scope readings in that case (for more discussion see Iatridou & Sichel, 2011; Lasnik, 1999)).

combination with a possibility modal.

- (118) $\neg\Diamond p$
- a. We can **in no way** approve the underlying directive.
 - b. This journey can **by no means** be said to have terminated if the Soul has to undergo yet another birth.
 - c. Eric could **hardly** go outside.
 - d. Anna can **rarely** borrow the car.
 - e. The ferry can carry **few** people.
 - f. She is doomed to continue longing **without** being allowed to hope.²¹

At least for some of these, the same impossibility reading can be found with a necessity modal:

- (119) $\Box\neg p$
- a. We must **in no way** approve the underlying directive.
 - b. This journey should **by no means** be said to have terminated if the Soul has to undergo yet another birth.
 - c. It should **hardly** be surprising that the service economy comes with numerous challenges.
 - d. The question of what is deserved should **rarely**, if ever, be asked.
 - e. The ferry must carry **few** people.

Negatively prefixed adjectives (like *impractical*) and negative adjectives like *hard* behave differently, in that the semantic negation they incorporate cannot take scope over modal verbs. Whereas (120a), with the standard negator *not* only has the impossibility reading, the examples (120b-c), with negative adjectives, only have an interpretation where the modal takes scope over the negation.²² Now earlier we saw that negative adjectives like *hard* license strong NPIs in their complement clause (Sect. 3.2). Taking this to be an indication for the presence of a logical negation inside such adjectives (as argued in De Clercq and Vanden Wyngaerd, 2019), one might expect this negation to take scope over a possibility modal, but this is not the case.

- (120) a. $\neg\Diamond p$ ($\equiv \Box\neg p$)
It **cannot** be easy to buy squid at Safeway on a Sunday.
- b. $\Diamond > \text{hard}$
It can be **hard** to buy squid at Safeway on a Sunday.
- c. $\Diamond > \text{impractical}$
It can be **impractical** to buy squid at Safeway on a Sunday.

Note that we did see NPIs in the complement clause of all these adjectives (Sect. 3.2 above), but these are clearly in a different structural position with respect to the negator than the modals in (120), i.e. while the NPIs are c-commanded by the negative adjectives, the modals are clearly c-commanding the negative adjectives.

The same pattern of behaviour is found with negative verbs like *deny* and *refuse* (121), as well as the negative preposition *without* (122).

²¹In nonfinite contexts, possibility modality cannot be realised by *can*, but needs the paraphrase *be allowed (to)*.

²²Predictably, the negation in negative adjectives also scopes under necessity modals like *must*, but in this respect they are no different from other negative elements, so we refrain from illustrating this.

- (121) $\Diamond >$ *refuse, deny*
 a. They could **refuse** to implement a cease-fire.
 b. She could **deny** her involvement in the assassination attempt.
- (122) $\Diamond >$ *without*
 He can walk **without** using a stick.

Note that these cases do not have to imply that there is no negation present in negative adjectives, negative verbs, and *without*, particularly in view of the fact that they are anti-additive and may license NPIs in their complement clause. The fact that we do find the impossibility reading when the modality is inside the complement clause to *without* confirms this idea (see (118f) above). What seems to be at issue here is that the negations in these adjectives and verbs cannot take scope in the clause that these modals belong to, a conclusion that will be reinforced by the evidence to be discussed in the next section.

We end with a brief note on squattive negators. Combined with the possibility modal *could*, they can take scope over or below the modal, in the same way as some of the negative indefinites discussed above (see also De Clercq, 2011 for this observation.) The following examples are due to William Harwood (p.c.).

- (123) a. $\neg\Diamond p$
 Helga could drink **fuck-all** whiskey, because Claudia had drunk the entire bottle on her own.
 ‘It was not possible for Helga to drink any whiskey.’
- b. $\Diamond\neg p$
 Claudia could drink **fuck-all** and give the booze to her friends, but she decided to down the bottle on her own.
 ‘It was possible for Claudia to drink nothing.’

The diagnostic for negation discussed in this section, the presence of impossibility readings with possibility modals, shows that most negative elements, apart from negatively prefixed adjectives, negative adjectives and negative verbs, interact with modal verbs in the same way as the standard negator does. In other words, the interaction with modality groups these negative words together, in spite of the fact that they can be split into an anti-additive and a merely downward entailing category. This diagnostic therefore shows that—across the anti-additive and downward entailing distinction—all of these words share a negative component, that surfaces in interaction with modal operators.

3.4 Syntactic effects of negation

In this section we look at a number of syntactic effects of negation. If all the anti-additive and downward entailing words and expressions discussed above were to contain a hidden negation, then we would expect to see syntactic effects of this. For a subgroup of these words that indeed turns out to be the case, though not for all of them. The syntactic effects we shall discuss will show the relevance of the scope of the negation. This scope may be limited in ways that are reflected in the syntactic effects.

Klima (1964) and McCawley (1998) investigated the negativity of most of the negative words and expressions that we discussed above. These authors have identified a series of diagnostic tests that are sensitive to the polarity of a sentence, i.e., to whether the sentence is affirmative or negative. One such test is the *tag question* test, which has the opposite polarity

to the sentence it tags onto. We call this a syntactic effect, since the polarity of the tag (positive or negative) does not affect the interpretation of the sentence at all, i.e. it is not semantic.²³

- (124) a. Hoboken is in New Jersey, **isn't** it?
b. Hoboken **isn't** in Pennsylvania, is it?

Another one is the *too/either* test, and the *so/neither* test, where each time the first member of the pair diagnoses an affirmative sentence, and the second member a negative sentence. This is illustrated in the examples below:

- (125) a. John voted for Bergland, and Mary voted for him **too**.
b. John **didn't** vote for Reagan, and Mary didn't vote for him **either**.
(126) a. John voted for Stassen, and **so** did Mary.
b. John **didn't** vote for Johnson, and **neither** did Mary.

In (125a) and (126a) the first clause of the coordination is affirmative, and we can get the adverbs *too* and *so* in the continuation, but not *(n)either*. Negative clauses like (125b) and (126b) show the mirror image of this: we get *(n)either* in the continuation, while *so* and *too* are excluded.

In what follows we will apply one of these tests, the question tag test, to the different types of negative words that we discussed in this chapter. We start by applying the question tag test to negative indefinites and the correlative conjunction *neither ... nor*. The examples in (127) show that these words tend to make a sentence negative, i.e. they occur with positive tags in the majority of cases.

- (127) a. **Nobody** knew the answer, did they?
b. The travel agents visited **no** Greek islands last summer, did they?
c. She knows **nothing** about the surprise party, does she?
d. They **never** go to the movies, do they?
e. There is **nowhere** to sit here, is there?
f. **Neither** Alissa **nor** Marian have read that book, have they?

Other anti-additive expressions like those in (128) also systematically trigger positive tags.

- (128) a. He is **by no means/in no way** a good father, is he?
b. Nasrin is **anything but** kind, is he?

However, downward entailing words predominantly trigger negative question tags, while positive tags are also sometimes allowed (see below for more discussion).

- (129) a. He can **hardly/barely/scarcely** move, %can he/can't he?
b. They **rarely/seldom** go to the movies, %do they/don't they?
c. **Few** hostages were released today, %were they/weren't they?
d. The prison released **few** hostages this month, %did it/didn't it?

Negatively prefixed adjectives on the other hand never trigger positive tags, and neither do negative adjectives (130).

²³Note that question tags need to be distinguished from reduplicative tags or same way tags. See Brasoveanu et al. (2014), De Clercq (2011), and Quirk et al. (1985) for discussion of the differences.

- (130) a. Your granddad is **unhappy**, isn't he?
 b. Your granddad is **sad**, isn't he?
 c. It's **hard** to find squid at Safeway, isn't it?

The same is true for the anti-additive word *without* (131), as well as sentences with negative verbs like *refuse* or *deny* (132):

- (131) He left **without** finding his keys, didn't he?
 (132) a. They **refused** to implement a cease-fire, didn't they?
 b. She **denied** any involvement in the plan, didn't she?

From these examples it seems that the question tags (and by extension the other syntactic tests) are sensitive to the scope of the negation. We can see this clearly in the following sentences from Payne (1985, p. 200) (also discussed in Penka, 2011, p. 5), in which the negator *not* occurs, which usually always makes the sentence negative, as in (133a), with a positive tag. But in (133b) we get an affirmative sentence because the negation takes scope under the adverb *often*, making the tag sensitive to *often* as it were, and not to the negation.

- (133) a. John **doesn't** often pay taxes, does he/*doesn't he?
 b. John often **doesn't** pay taxes, *does he/doesn't he?

In the sentences of (129), too, the possibility of the different tags can be explained, at least in part, in terms of negation taking sentential scope or not. For example, one can establish that *few* behaves differently in subject position and in object position, as shown in an experimental study by Brasoveanu et al. (2014). In this study, participants in an experiment had to select a question tag (positive or negative) for a stimulus. The results are shown in Table 3 (adapted from Brasoveanu et al., 2014, p. 180).

Table 3: Stimuli and tags, in ascending order of the frequency of the negative tag

STIMULUS	NEG		POS
	N	%	N
negative indefinite subject	60	21%	223
negative indefinite adverb	71	24%	221
<i>not</i>	75	26%	214
negative indefinite object	100	34%	186
<i>rarely, scarcely, hardly</i>	170	59%	119
<i>few</i> subject	183	64%	103
<i>few</i> object	241	84%	45
affirmative sentence	275	95%	14

The relevance of scope is obvious from the contrast between negative indefinites as subjects (which trigger only 21% negative tags) and as objects (34% negative tags). With *few* in subject position, a negative tag is chosen in 64% of the cases; with *few* in object position, this rises to 84%. In other words, a negative indefinite or *few* in object position has a much harder time taking sentence scope, and thus making the sentence negative. The case of the *un*-prefix in (130) can also be explained in these terms, in the sense that *un*- can never have sentence

scope. The same goes for *without* in (131): it never makes the containing sentence negative, though clearly its complement clause is negative in all relevant respects.

Now when we look at squative negation we get a completely different picture: in subject and in object position squative negation can only license negative tags, as illustrated in (134).

- (134) a. Janet read **squat** didn't she? (Postal, 2004, p. 164)
b. Janet read **fuck-all** books, didn't she?
c. **Fuck-all** men love her, don't they? (De Clercq, 2011, pp. 15–16)

If the question tag test is a syntactic test for negation, then these data suggest that squative negation is either syntactically non-negative, or that the negation inside these elements is not able to scope high enough. We will get back to this in section 3.5.

3.5 Discussion

The results of the diagnostic tests for negation that we discussed in the previous sections are summarized in Table 4. In the rows we find the diagnostics that we have used, in the columns the different types of negative elements: the standard negator (SN), negative correlative conjunctions (NCC), negative indefinites and other negators of the anti-additive type (NI), negative adjectives, verbs, and *without* (NAVW), downward entailing (non-anti-additive) negative quantifiers (DE), squative negators of the type discussed by Postal (2004) (SQ), and the British English *fuck-all* (FA). The question mark indicates mixed behaviour, for example the fact that negative indefinites sometimes do, and sometimes do not, trigger positive tags, depending on their syntactic position.

Table 4: Overview of the diagnostics for negation

	SN	NCC	NI	NAVW	DE	SQ	FA
De Morgan's Law I	✓	✓	✓	✓	✗	✓	✓
NPI-licensing (strong NPI)	✓	✓	✓	✓	✗	✗	✓
NPI-licensing (weak NPI)	✓	✓	✓	✓	✓	✗	✓
Impossibility readings	✓	✓	✓	✗	✓	✓	✓
Positive question tags	✓	✓	?	✗	?	✗	✗

With the exception of squative items (see below), Table 4 clearly shows that De Morgan's Law I and strong NPI-licensing similarly distinguish the anti-additive from the merely downward entailing negative words. The property of the impossibility readings with possibility modals on the other hand unites these two groups again by being equally sensitive to their underlying negative component. That property isolates another group, that of the negative verbal and adjectival predicates and *without* (NAVW), which are the only group that never gives rise to impossibility readings. The reason for this seems to be structural in nature: the negative predicate is in most cases contained within a modal complement, and cannot scope into the matrix clause from that position.

The syntactic question tag test discussed in Sect. 3.4 clearly stands out as a diagnostic, yielding results that are rather different from those of the other diagnostics. It indicates that there are not only different types of negation with different scopes, but also that identical forms of negation, like the standard negator in (133), can have different scopes. The sentences that are negative according to the question tag test are those where the negation has scope

over the entire sentence, i.e. where negation takes widest scope, and hence types the clause as negative. At first sight, one might be inclined to think that the syntactic tests are sensitive to some type of external negation, as discussed in Sect. 2.1.1. However, the tests could also be sensitive to a form of internal negation. Internal negation has been claimed to be realisable in different structural positions in the clause (e.g. in work by Cormack & Smith, 2002; Ouhalla, 1990; Zanuttini, 1997). One of these could be high up in the clausal structure, i.e. in the CP domain, from where negative clause typing happens (De Clercq et al., 2012; Laka, 1990; Moscati, 2006; Progovac, 1993). This could then also be the position to which question tags are sensitive: if this position gets activated by a negative feature (via whichever mechanism, i.e. base generation of a negator, Agree, movement, or QR), this results in positive question tags. However, whenever this high position cannot be overtly or covertly activated, the syntactic tests will give the impression that the sentence is positive (by means of negative tags for instance).

However, being syntactically positive clearly does not rule out the presence of a negation within the clause, as we saw with the other diagnostics. The reasons for a negation to not be able to type the clause as negative can follow from a variety of different reasons. One of these reasons can be that the negative feature is too deeply embedded within the structure to type the clause as negative (as for instance with negative indefinite objects, or with DE quantifiers) (see for instance De Clercq et al., 2012, for an analysis). Another reason can be that there is an intervening positive quantifier preventing the negation from typing the clause as negative, as we saw in (133) (see also De Clercq, 2013, for discussion). A third possible reason has to do with the internal syntactic structure of the negative word: under the assumption that the negator does not only consist of a semantic $\neg$ with a corresponding syntactic [NEG], but also of scope-related syntactic features, the absence of certain scope features can prevent the negator from taking widest scope as well (as in the nanosyntactic proposal by De Clercq, 2013, 2020; De Clercq & Vanden Wyngaerd, 2019). This can account for why negations contained within negative verbs or (negatively prefixed) adjectives systematically cannot give rise to positive question tags (i.e. negative sentences) or to impossibility readings with modals.

Finally, we need to say something about English squattive items. On the one hand they do not give rise to positive question tags, regardless of where they appear in the clause. That puts them at first sight in the same box as negative adjectives and verbs. However, we cannot claim that they consist of a similar low scope negation, because - unlike negative verbs or adjectives - they interact with modal verbs in the same way as other negative indefinites, i.e. they take wide scope. For that reason De Clercq (2011) proposes that squattive negation is semantically negative (i.e. anti-additive or downward entailing), but does not contain any syntactic negative features, hence leading to positive clause typing. In other words, it seems that a word can be semantically or conceptually negative, without having any corresponding negative features in syntax (cf. also Buschbom et al., 2024; Zeijlstra, 2008).

The diagnostics show that De Morgan's Law I, NPI licensing, and the interaction with modality track the presence of a semantic negation inside the relevant morphemes, while the question tag test (and the other Klima tests) is sensitive to the presence of a syntactic negative feature with wide scope. While the presence of a syntactic negative feature seems to always coincide with an underlying semantic negation²⁴, the opposite is not the case,

²⁴We abstract away here from cases of negative concord for which it has been argued in one part of the literature that it arises as a consequence of pure syntactic negativity (Zeijlstra 2004a). However, as discussed in section 2.2.2 there are multiple analyses for negative concord, including analyses that consider negative indefinites as semantically negative quantifiers or as NPIs. The latter analyses would be compatible with the claim made in this conclusion.

as shown by the squative negators, which are semantically/conceptually negative but not syntactically negative. It would be interesting to investigate whether the distinction between negators that are both syntactically and semantically negative and negators that are only semantically negative is reflected in terms of processing time (see for instance Clark & Chase, 1972; Dudschig et al., 2019; Tian et al., 2010, for relevant literature on processing negation). If there turns out to be a difference between these type of negators, then this may shed further light on how negation is cognitively anchored.

4 Conclusion

In this chapter, we have first classified negative elements in natural language into five different broad categories: standard negators, negative correlative conjunctions, negative indefinites, squative negators, and other types of negative elements. We next discussed a number of criteria by which to diagnose negation, such as the universally negative reading of *or* in the presence of negation (based on De Morgan's Law), the licensing of Negative Polarity Items (NPIs), the interaction with modality, and the syntactic effects of negation. We concluded that the diagnostics point to the fact that the distinction between semantic and syntactic negation is real, and that not all semantically negative words come with a syntactic negative feature, while the opposite seems always the case.

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